

PUCHALAPALLI HARIKA

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SUMMARY

AI Engineer experienced in the full ML lifecycle (data preparation, training, tuning, retraining) and in deploying LLM/RAG agents; full-stack with Python, PyTorch, FastAPI, and React/TypeScript.

EDUCATION

Mahindra University, Hyderabad – B.Tech in Artificial Intelligence CGPA: 9.30/10	Aug 2023 – Aug 2027
Sri Venkateswara Junior College, Visakhapatnam – Intermediate (MPC): 94.5%	April 2021 – May 2023
The Presidential School, Visakhapatnam – Class X (CBSE): 95.8%	April 2020 – April 2021

EXPERIENCE

AI Engineering Intern, Centific <i>Offer earned by winning Centific's hackathon</i> - Built a Market Research Agent that automates multi-step market analysis using LangGraph, cutting manual research effort - Implemented persistent conversation history with Cosmos DB so analysts can pause and resume research threads across sessions - Designed Azure event-driven processing (Azure Event Hub) to run long analysis steps asynchronously and reliably	Jun 2026 – Sept 2026
AI Engineering Intern, Vassar Labs - Built and deployed an end-to-end RAG-based QA system covering document ingestion, chunking, embeddings, semantic retrieval, and RESTful APIs - Integrated Milvus DB, containerized services with Docker and docker-compose, and validated all endpoints using Postman	Dec 2025 – Jan 2026

PROJECTS

Aegis: Multi-Layer, Explainable Fraud Detection Platform 🔄 - Designed a two-layer detection pipeline: XGBoost for known fraud patterns plus Isolation Forest for novel anomalies, reaching PR-AUC 0.97 on imbalanced data with calibrated fraud probabilities - Built a RAG-based LangGraph investigation agent that retrieves similar past cases from pgvector, reasons conditionally, and uses SHAP to explain in plain language why each transaction was flagged - Enabled privacy-preserving cross-institution signal sharing with salted cryptographic hashes, letting banks share fraud signals without exposing customer data - Built the investigator dashboard in React + TypeScript to review alerts, explanations, and agent findings Tech Stack: Python, XGBoost, Isolation Forest, SHAP, LangGraph, PostgreSQL/pgvector, React, TypeScript	Aug 2026 – Sept 2026
EduAgent: Personalized Agentic AI Tutoring System 🔄 - Built an adaptive AI tutor for AI/ML learners that remembers each student's level, weak concepts, and progress across sessions Fine-tuned DistilBERT to classify queries as beginner/intermediate/advanced (97.92% accuracy), routing each learner to the right teaching depth - Implemented two-pass personalized RAG with MiniLM embeddings, retrieving both relevant content and the learner's weak concepts; used Llama-3.3-70B (Groq) as an evaluator to track mastery Tech Stack: Python, DistilBERT, Sentence-Transformers (MiniLM), Llama-3.3-70B, Groq, SQLite	Mar 2026 – May 2026
Unified Multilingual NLP Model for Indian Languages 🔄 - Built one unified model for three tasks: transliteration (22 languages), text normalization (12), and punctuation restoration (23) - Created a large, high-quality multilingual dataset by combining real native-language data with Gemini 2.0 Flash synthetic data Supervised fine-tuned Gemma-3-1B-PT with multi-task instruction learning; benchmarked against Sarvam on BLEU, CER, WER, and chrF++ Tech Stack: Python, PyTorch, Hugging Face Transformers, Gemma-3, Gemini 2.0 Flash	Oct 2025 – Dec 2025
Hybrid Predictive Maintenance, NASA CMAPSS 🔄 - Converted unlabeled engine sensor time-series into health-stage labels using weighted K-Means and PCA - Trained classifiers and regressors to predict health stage and Remaining Useful Life, combining both into a risk score for proactive alerts Tech Stack: Python, Scikit-learn, K-Means, PCA, XGBoost, Random Forest	Mar 2025 – May 2025
Medical Scribe Agent (In Progress) 🔄 Currently developing an AI medical scribe agent, extending my Centific hackathon-winning solution, that converts doctor-patient conversations into structured clinical notes Tech Stack: Python, LangGraph, LLMs	Jun 2026 – Present

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, SQL
ML Lifecycle: Data Preprocessing, Feature Engineering, Model Training, Hyperparameter Tuning, Cross-Validation, Evaluation, Retraining, Fine-Tuning
AI / GenAI: Deep Learning, NLP, LLMs, RAG, Agentic AI, LangGraph, Embeddings, Vector Search, SHAP, Prompt Engineering
Libraries: PyTorch, Hugging Face Transformers, Sentence-Transformers, Scikit-learn, XGBoost, NumPy, Pandas
Web & Tools: React, FastAPI, Node.js, REST APIs, HTML/CSS, PostgreSQL/pgvector, SQLite, Milvus, Docker, Git, Postman, Azure (Cosmos DB, Event Hub)

AWARDS & ACHIEVEMENTS

- Winner, Centific Hackathon:** ranked **1st among 15,000+ applicants** in a 14-day hackathon for a medical scribe agent; won an internship May 2026
- Merit Scholarship** (2 × INR 1,00,000), Mahindra University, for academic excellence April 2023 – April 2025

LEADERSHIP & ACTIVITIES

Finance Member, Sakhi Club (Women Empowerment Club), Mahindra University Managed club finances and budgets, and coordinated awareness initiatives supporting women empowerment programs	Aug 2025 – Present
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